What Will you learn in This Masterclass?

- Introduction of **Power Pivot**
- Why We use **Power Pivot?**
- How to Activate **Power Pivot Tab** in Excel?
- We will work with **real business data**
- How to build **Relationship between Table?**
- We will learn How to **analyze data**
- We will do **time-based comparisons**
- Dax Formulas **step-by-step**
- **Beginner to advanced**—no skipping

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What is Power Pivot?

Power Pivot is an advanced data modeling and analysis engine inside Microsoft Excel.

Power Pivot helps Excel handle **BIG data**, **multiple tables**, and **complex calculations**—very fast and very easily.

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Normally in Excel:

- We work with **one table**
- We use Multiple **VLOOKUP / XLOOKUP -** slow files
- We use **Pivot Tables** work on **single flat tables**
- Excel becomes **slow -** Complex formulas become unmanageable
- Only **1 million rows limit,** File becomes **heavy**
- Performance issues with large datasets

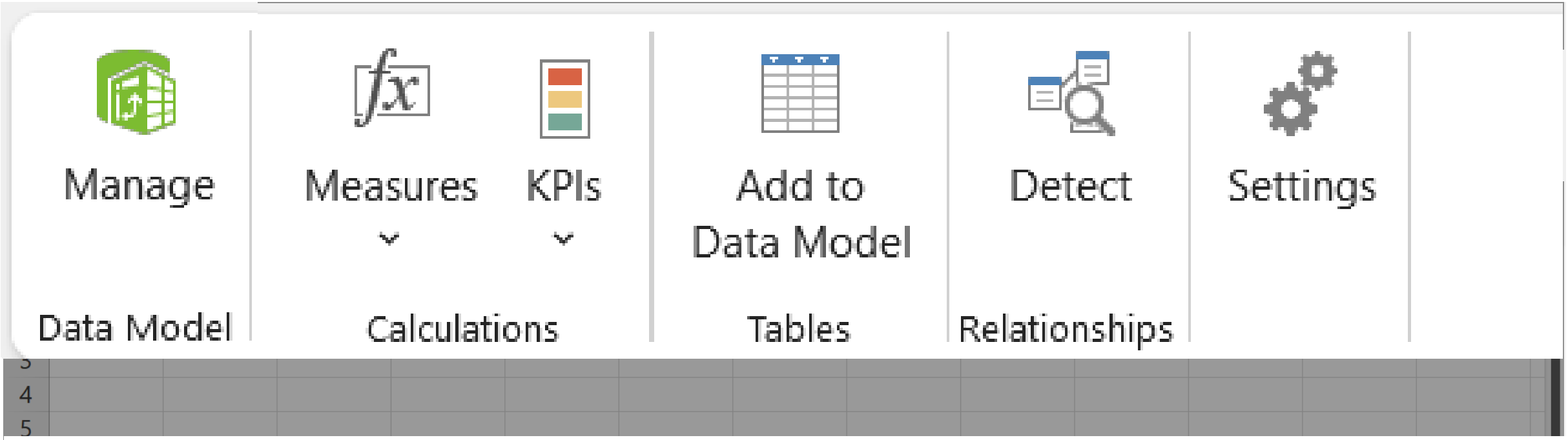
Microsoft realized one thing clearly:

Excel users need **database-level power** but inside Excel itself.

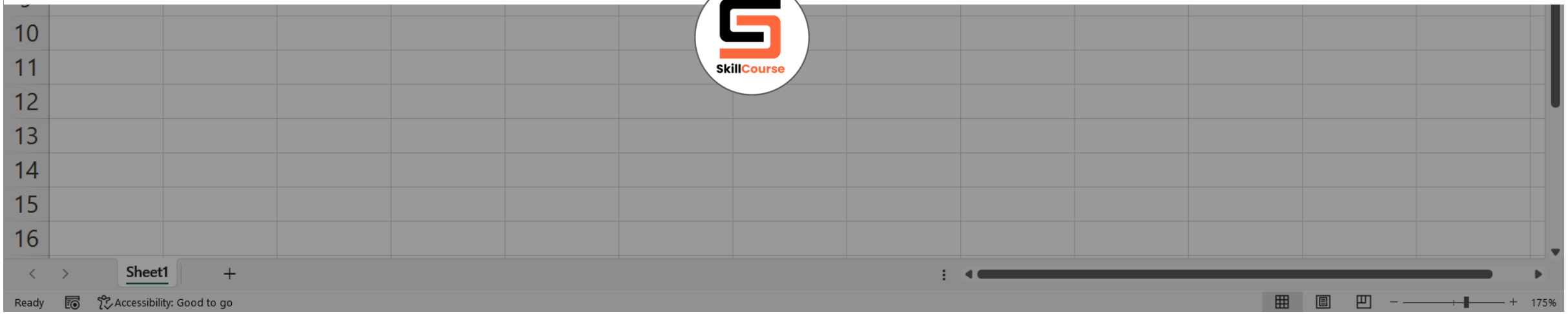
Power Pivot **solves** all these problems.

One-line definition

Power Pivot is an Excel tool that allows us to **load large data**, **create relationships** between multiple tables, and perform **powerful calculations** using DAX.



How to activate Power Pivot Tab in Excel?



What Is the Real Power Behind Power Pivot?

In-Memory Engine (VertiPaq)

Power Pivot uses **VertiPaq**, a **column-based in-memory engine**

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What does that mean?

Data is **compressed**

Data is **stored in RAM**

Calculations are **extremely fast**

Even millions of rows work smoothly

Why Should We Use Power Pivot?

1. Handle Large Data Easily
2. Multiple Tables – No More Lookups
3. Real Data Modeling (Like Power BI)
4. Powerful Calculations Using DAX
5. Clean & Professional Reports

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Who Should Learn Power Pivot?

- Data Analysts
- MIS Executives
- Excel Professionals
- Finance & Accounts Professionals
- Sales & Marketing Analysts
- Anyone moving towards **Power BI**

How to Enable Power Pivot in Excel (Step-by-Step)

Step 1	Open Excel → Open Microsoft Excel – open blank workbook
Step 2	Go to Options → Click File - Click Options
Step 3	Add-ins → Go to Add-ins - At the bottom: Manage: COM Add-ins - Click Go
Step 4	Enable Power Pivot → Tick ✅ Microsoft Power Pivot for Excel - Click OK
Step 5	Verify - You will see a new tab: Power Pivot in Excel Ribbon

“From the next chapter, we will stop talking theory and start working with real data.”

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Chapter 2 : Importing Data
& Create relationships

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Chapter Goal

- What is our dataset (tables + purpose)
- How to **import multiple CSV files into Power Pivot**
- How to **create relationships**
- How to **verify relationships are working**

Dataset Story

Assume this is a retail company selling products across multiple Indian states using **stores + online hubs.**

We have one **full year of sales**, and we want to **analyze monthly** trends, **category** performance, **customer** segments, **discounts, returns**, **profit**, and many more insights.

Table Structure

Customers.csv	Date.csv	Products.csv	Sales_Fact.csv	Stores.csv
(2,500 rows)	(365 rows)	(180 rows)	(20,000 rows)	(45 rows)

Fact Table (Main)

- **Sales_Fact.csv** (Transaction table)
  - Each row = one order line
  - Contains keys to connect with other tables

Dimension Tables (Support tables)

- **Date.csv**
  - DateKey (primary key)
  - year, month, quarter, fiscal year, weekend, etc. This is the backbone for time intelligence.
- **Products.csv**
  - product details + category/subcategory/brand
  - pricing + cost
- **Customers.csv**
  - segment, city, state, customer since, preferred channel
- **Stores.csv**
  - region/state/city + channel (Retail/Online Hub/B2B Desk)

Relationships

Create these relationships (One-to-Many):

- Date[DateKey] → Sales_Fact[OrderDateKey]
- Products[ProductID] → Sales_Fact[ProductID]
- Customers[CustomerID] → Sales_Fact[CustomerID]
- Stores[StoreID] → Sales_Fact[StoreID]

Customers.csv

Date.csv

Products.csv

Sales_Fact.csv

Stores.csv

How to Import CSV into Power Pivot

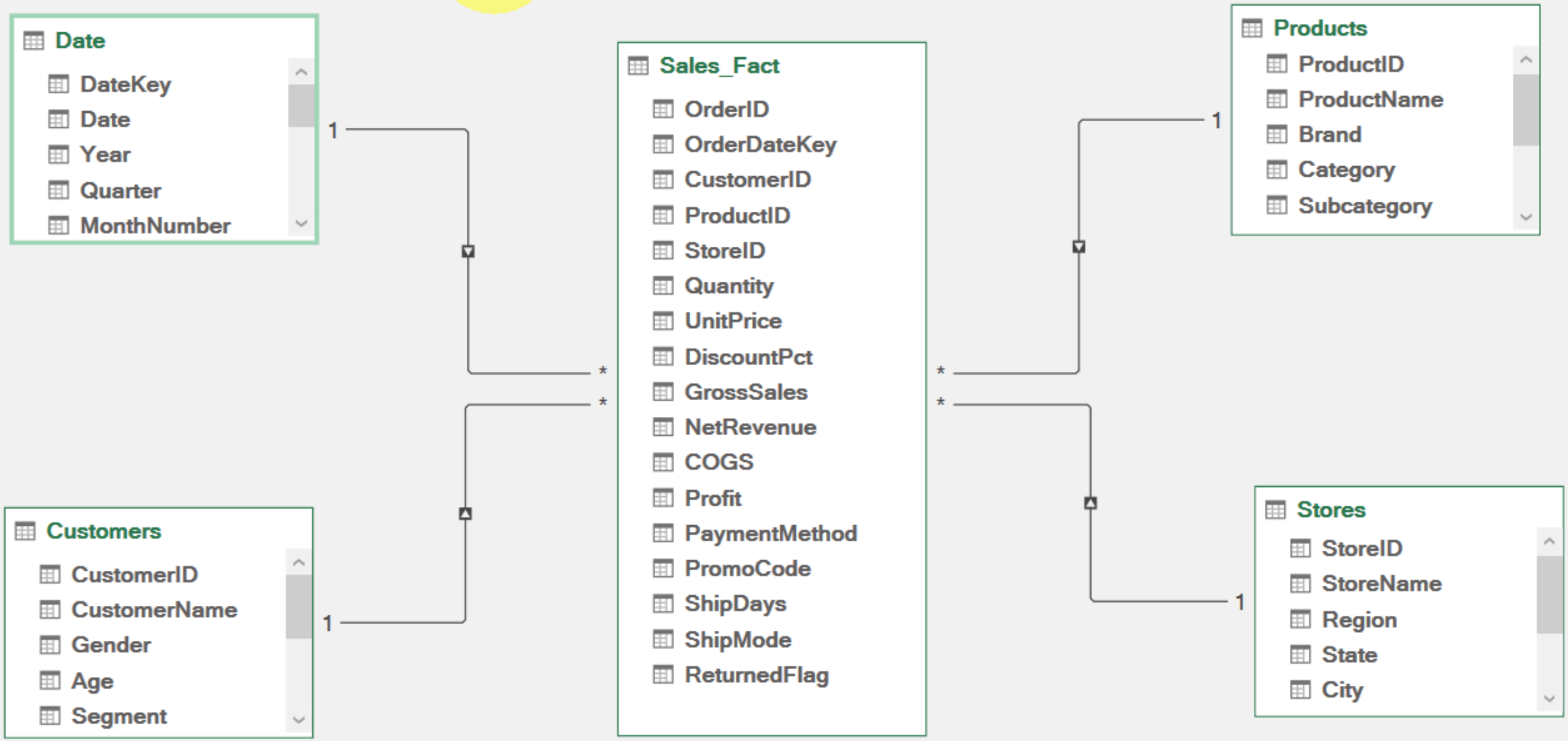
- Excel → **Data** tab
- **Get Data → From File → From Text/CSV**
- Select the CSV (Example: Sales_Fact.csv)
- In preview window, click **Load To...**
- Choose ✅ **Only Create Connection**
- Tick ✅ **Add this data to the Data Model**
- Click **OK**

- Repeat this for **each CSV file**:

How to Create Relationships

- Power Pivot → **Manage**
- Go to **Diagram View**
- Drag & drop:
  - Date[DateKey] to Sales_Fact[OrderDateKey]
  - Products[ProductID] to Sales_Fact[ProductID]
  - Customers[CustomerID] to Sales_Fact[CustomerID]
  - Stores[StoreID] to Sales_Fact[StoreID]

    OR
- Power Pivot → **Manage** → **Design** → **Manage Relationships** → **New**


Date
DateKey
Date
Year
Quarter
MonthNumber
Customers
CustomerID
CustomerName
Gender
Age
Segment
Sales_Fact
OrderID
OrderDateKey
CustomerID
ProductID
StoreID
Quantity
UnitPrice
DiscountPct
GrossSales
NetRevenue
COGS
Profit
PaymentMethod
PromoCode
ShipDays
ShipMode
ReturnedFlag
Products
ProductID
ProductName
Brand
Category
Subcategory
Stores
StoreID
StoreName
Region
State
City
1
*
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1
1
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1

How to Verify Relationship is Working

- **Test #: Create Pivot from Data Model**
- Insert → PivotTable
- Choose ✅ **Use this workbook's Data Model**
- Put:
  - Rows: Date[MonthName]
  - Values: Sales_Fact[NetRevenue]

- If you see month-wise revenue → relationship is working ✅

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Chapter 3 - **DAX** Fundamentals + Business **KPIs**

Goal

- Understand **what DAX is**
- Know **why DAX is different from Excel formulas**
- Understand **Measures vs Calculated Columns**
- Write **core DAX formulas confidently**
- Create **real business KPIs**

Fear of DAX

If you are **scared** of **DAX**, you are not alone.

Every **Power BI** expert was once afraid of **DAX**.

The good news? **DAX is NOT difficult** — it is **logical.**

Important

DAX is not about **memorizing formulas**.

DAX is about **understanding** business **logic**.

What is DAX?

DAX = Data **Analysis** Expressions

- DAX is a formula language used to perform calculations on data models.

Used in:

- Power Pivot
- Power BI
- SSAS Tabular

One-line definition:

- **DAX helps us calculate business metrics on top of related tables.**

DAX vs Excel Formulas (Very Important Slide)

Excel Formula	DAX
Works on cells	Works on tables
Row-by-row	Context-based
Sheet-based	Data Model-based
SUM(A1:A10)	SUM(Sales[NetRevenue])
No relationships	Uses relationships

Excel = **Cell** thinking

&

DAX = **Business** thinking

Two Types of DAX

Calculated Column

- Created **inside a table**
- Calculated **row by row**
- Stored in memory
- Used for classification

Example :

- Sales[SalesAmount] = Sales[Quantity] * Sales[UnitPrice]

Dax Type

Measure (MOST IMPORTANT)

- Calculated **at runtime**
- Responds to filters
- Used in KPIs, Pivot Tables, Dashboards
- No extra memory usage

Example:

- Total Sales = SUM(Sales_Fact[NetRevenue])

Where to Write DAX in Power Pivot

1. Power Pivot → **Manage**
2. Choose **Sales_Fact** table
3. Click **Calculation Area**
4. Write Measure
5. Press Enter

Core DAX Functions (Foundation)

SUM

- Total Sales = SUM(Sales_Fact[NetRevenue])

COUNTROWS

- Total Orders = COUNTROWS(Sales_Fact)

DISTINCTCOUNT

- Total Customers = DISTINCTCOUNT(Sales_Fact[CustomerID])

AVERAGE

- Average Discount = AVERAGE(Sales_Fact[DiscountPct])

Business KPIs

- **KPI 1: Total Sales**
  - Total Sales = SUM(Sales_Fact[NetRevenue])
- **KPI 2: Total Profit**
  - Total Profit = SUM(Sales_Fact[Profit])
- **KPI 3: Total Orders**
  - Total Orders = COUNTROWS(Sales_Fact)
- **KPI 4: Average Order Value (AOV)**
  - Total Sales ÷ Total Orders
  - AOV = DIVIDE([Total Sales],[Total Orders])
- **KPI 5: Profit Margin %**
  - Profit Margin % = DIVIDE([Total Profit],[Total Sales])

Understanding Filter Context

Filter context means

- **“What data is currently visible?”**

Examples:

- Month selected → only that month data
- Category selected → only that category data
- Customer selected → only that customer data

DAX automatically reacts to filters.

- **This is the magic of DAX.**

CALCULATE – The Heart of DAX

If DAX has a heart,

that heart is **CALCULATE()**

What CALCULATE does:

Changes filter context

Applies conditions

Creates intelligent measures

Syntax:

CALCULATE(Expression,Filter1,Filter2)

Filter using Calculate

Electronics Sales

=CALCULATE([Total Sales],Products[Category] = "Electronics")

Online Sales

=CALCULATE([Total Sales],Stores[Channel] = "Online Hub")

RELATED Function

RELATED pulls data from **one-side table to many-side table**

Example: Product Category = RELATED(Products[Category])

Works only in **Calculated Columns**, not Measures.

Calculated Column

IF & Logical DAX

Example: Profit Status

```
Profit Status =IF(
Sales_Fact[Profit] >
0,"Profit","Loss")
```

KPI: Return Rate (%)

Returned Orders ÷ Total Orders

```
Return Rate % =
DIVIDE(
   [Returned Orders],
   [Total Orders]
)
```

KPI: Customer Segmentation Insight

```
Consumer Sales =
CALCULATE(
    [Total Sales],
    Customers[Segment] = "Consumer"
)
```

Common Beginner Mistakes

- ❌ Writing Measures as Columns
- ❌ Using = instead of == logic thinking
- ❌ Ignoring relationships
- ❌ Overusing calculated columns
- ❌ Fear of CALCULATE

“DAX is **logical**, not **magical**.”

you didn’t just learn formulas.
You learned how **businesses** think with **data**.”

Chapter 4 **Time Intelligence** DAX

What is Time Intelligence?

Time Intelligence means **comparing data across time**.

This month vs last month

This year vs last year

YTD sales

Trend analysis

Every business decision depends on time comparison.

Why Date Table is Mandatory

“Without a proper Date table, DAX Time Intelligence will FAIL.”

- **Why?**
- Sales table has repeated dates
- DAX needs **continuous dates**
- Functions like YTD, QTD, MTD need a calendar

- **One Golden Rule**
- Date table must be related to Sales table
- Date[DateKey] → Sales_Fact[OrderDateKey]

Year-To-Date (YTD)

Total sales from start of the year till today

```
Sales YTD =
TOTALYTD(
   [Total Sales],
   Date[Date]
)
```

Month-To-Date (MTD)

Sales from start of current month till today

```
Sales MTD =
TOTALMTD(
   [Total Sales],
   Date[Date]
)
```

Quarter-To-Date (QTD)

Sales from start of current Quarter till today

```
Sales QTD =
TOTALQTD(
   [Total Sales],
   Date[Date]
)
```

Chapter 5 Advanced **Date** and **Time Intelligence** DAX

Last Year Sales (LY)

“How much did we sell last year for the same period?”

```
Sales Last Year =
CALCULATE(
   [Total Sales],
   SAMEPERIODLASTYEAR(Date[Date])
)
```

Previous Month Sales

```
Sales Previous Month =
CALCULATE(
   [Total Sales],
   PREVIOUSMONTH(Date[Date])
)
```

Common Time Intelligence Mistakes (Must Warn Students)

- ❌ No Date table
- ❌ Using text dates
- ❌ Wrong relationship direction
- ❌ Writing time logic on Sales table
- ❌ Using columns instead of measures

“Time Intelligence works only when modeling is correct.”

“Now you are no longer just calculating numbers.
You are analyzing business performance over time.”